

CHE 321 - Exam 2 - Sample 1

exam2.m

```
1 function h = PFRwHEAT_fun(V,y)
2 FA = y(1);
3 FB = y(2);
4 FC = y(3);
5 FD = y(4);
6 T = y(5);
7 Ta = y(6);
8
9
10 FT = FA + FB + FC + FD;
11 T0 = 300;
12 k1A = 40;
13 k2C = 2;
14 kC = 0.5;
15 R = 1.987;
16 E1 = 8000;
17 E2 = 12000;
18 mCool = 50;
19 Ua = 80;
20 Ta0 = 325;
21 FA0 = 5;
22 FB0 = 10;
23 FC0 = 0;
24 FD0 = 0;
25 FT0 = FB0 + FA0 + FC0 + FD0;
26 CT0 = 0.2;
27 v0 = FT0/CT0;
28 v = v0*(FT/FT0)*(T/T0);
29
30
31 k1 = k1A*exp((E1/R) * ((T0)^-1 - (T)^-1)); %I forgot to include the arhenius relation on my paper
32 k2 = k2C*exp((E2/R) * ((T0)^-1 - (T)^-1));
33
34
35 CpA = 10;
36 CpB = 10;
37 CpC = 30;
38 CpD = 20;
39 CpCool = 10;
40
41
42 CA = CT0*(FA/FT)*(T0/T);
43 CB = CT0*(FB/FT)*(T0/T);
44 CC = CT0*(FC/FT)*(T0/T);
45 CD = CT0*(FD/FT)*(T0/T);
46
47
48 r1A = -1*k1*CA*CB^2;
49 r2A = -1*k2*CA*CC;
50 rA = r1A + r2A;
51
52 r1B = -2*k1*CA*CB^2;
53 rB = r1B;
54
55 r1C = k1*CA*CB^2;
56 r2C = -1*k2*CA*CC;
57 rC = r1C + r2C;
58 RC = kC*CC;
59
60 r2D = k2*CA*CC;
61 rD = r2D;
62
63
64 deltaH1A = -30,000;
65 deltaH2A = -10,000;
66 Ua = 80;
67
68
69 h1 = rA;
70 h2 = rB;
71 h3 = rC - RC;
72 h4 = rD;
73 h5 = ( (r1A*deltaH1A + r2A*deltaH2A) - Ua*(T-Ta) ) / (FA*CpA + FB*CpB + FC*CpC + FD*CpD);
74 h6 = (Ua*(Ta - T)) / (mCool*CpCool); %I forgot to account for the heat exchanger fluid being countercurrent on my paper
```

```

75
76
77 h = [h1; h2; h3; h4; h5; h6];
78 end
79
80
81
82 clc; clear all;
83
84
85 Vspan = [0 10];
86 Y0 = [5 10 0 0 300 325];
87
88 [V,y] = ode45(@PFRwHEAT_fun,Vspan,Y0);
89
90 FA = y(:,1);
91 FB = y(:,2);
92 FC = y(:,3);
93 FD = y(:,4);
94 T = y(:,5);
95 Ta = y(:,6);
96
97 FA0 = 5;
98 FB0 = 10;
99 FC0 = 0;
100 FD0 = 0;
101 FT0 = FB0 + FA0 + FC0 + FD0;
102 CT0 = 0.2;
103 T0 = 300;
104 FT = FA + FB + FC + FD;
105 v0 = FT0/CT0;
106 v = v0.*(FT/FT0).*(T/T0);
107
108
109 CA = FA./v;
110 CB = FB./v;
111 CC = FC./v;
112 CD = FD./v;
113
114
115 figure(1)
116 plot(V,FA);
117 hold on
118 plot(V,FB);
119 plot(V,FC);
120 plot(V,FD);
121 title('Molar Flow Rates')
122 xlabel('PFR Volume(dm^3)')
123 ylabel('Flow Rates(mol/s)')
124
125 figure(2)
126 plot(V,T);
127 hold on
128 plot(V,Ta);
129 title('T and Ta')
130 xlabel('PFR Volume(dm^3)')
131 ylabel('Temperatures(K)')
132
133 figure(3)
134 plot(V,CA);
135 hold on
136 plot(V,CB);
137 plot(V,CC);
138 plot(V,CD);
139 title('Concentrations')
140 xlabel('PFR Volume(dm^3)')
141 ylabel('Concentrations(mol/dm^3)')

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